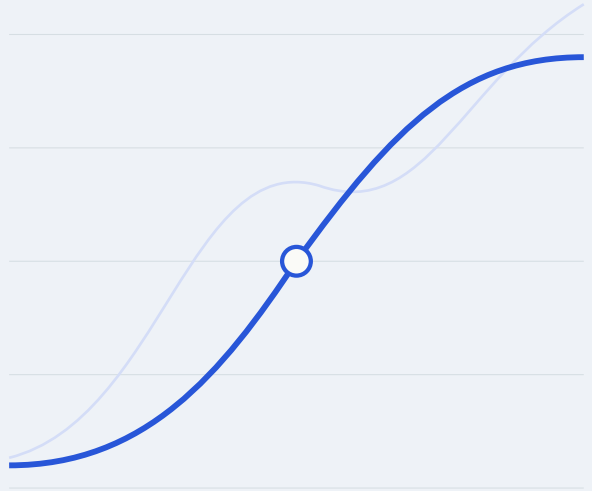
# Polymarket: Turning Future Events into Tradable Assets

A Guide for Economics & Finance Researchers




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# Real events become tradable probabilities

- A prediction market converts the outcome of a real-world event into tradable shares across politics, sports, economics, technology, and weather.
- Each outcome share trades in [0, 1] USD and is read directly as the market's implied probability.

  Why [0,1]? Each share is backed by $1 of collateral, which pins the price into this range – derived on the next slide.
- This is not "betting": shares map to real outcomes and pay off via a resolution rule, not a bookmaker.
- Settlement is on-chain; results are certified by an oracle, improving transparency.

  "On-chain" = recorded on a public blockchain. Here: the Polygon network, using USDC, a dollar-pegged stablecoin.

**Minimal example: a binary market**

Question: "Will candidate X win the election?" [ YES ] [ NO ]

Buy YES @ 0.60; if it happens → redeem 1.00 (profit 0.40); if not → YES goes to 0.

Scope: this deck simplifies to binary YES/NO markets; multi-outcome markets use the same collateral logic.

# One YES + one NO are backed by $1

- Built on the Gnosis Conditional Token Framework – an open smart-contract standard that mints YES/NO tokens and locks $1 of collateral per pair.
- Every YES + NO pair is backed by $1 of collateral – the "fully collateralized" promise.
- Users can split $1 into 1 YES + 1 NO, or merge 1 YES + 1 NO back into $1.
- The identity guarantees that payouts sum to $1 regardless of the outcome.

**FULL COLLATERAL / SPLIT AND MERGE**

**1 YES** + **1 NO** ⇄ **$1 USDC**

Split: $1 becomes a token pair. Merge: a token pair returns $1.

# Two ways to acquire a YES share

**Method 1 · Secondary trade (buy from a holder)**

You post "buy 1 YES @0.60"; if a holder sells, orders match and existing tokens change hands – no new collateral is created.

**Method 2 · Co-minting (paired with a NO buyer)**

You buy YES@0.60 while another buys NO@0.40; the two payments sum to exactly $1, and the system mints a fresh YES/NO pair: YES to you, NO to them.
Matching: the engine first fills against resting orders (Method 1); only unmatched flow mints new pairs (Method 2).

# The order book determines the price

- User limit orders form the Central Limit Order Book (CLOB).
- The displayed price is usually the midpoint; when the spread exceeds $0.10, the last trade is shown.
- Trader order flow drives price discovery; the platform matches and clears.
- Thin liquidity widens spreads and increases reliance on the last trade.

A limit order rests at a set price. The bid is the highest buy price; the ask is the lowest sell price.

| ASK / SELL | |
|---|---|
| 0.62 | 1,200 |
| 0.61 | 3,500 |
| **0.60** | 5,000 |
| **BID / BUY** | |
| **0.59** | 4,200 |
| 0.58 | 2,800 |
| 0.57 | 1,100 |

**READING THE BOOK**

Spread = 0.60 − 0.59 = 0.01  Midpoint = (0.60 + 0.59) / 2 = 0.595

# Market makers supply liquidity and bear risk

- Without resting orders, a buyer cannot fill instantly. Market makers continuously post bids and asks.
- Typical quote: YES bid = 0.59, ask = 0.61 – earn the spread by buying low and selling high.
- But MMs bear inventory risk (wrong-way exposure in trends) and adverse selection (informed traders trade against them).
- In a one-sided YES rally, the MM may be forced into a short YES position – not a sure win.

> MM posts orders (bid 0.59 / ask 0.61) ↓ earn spread − inventory / adverse risk

# MM Profit = Spread + Rewards − Risk

- On the surface, three revenue streams: spread + rebates + liquidity incentives.

- True profit must net out inventory risk and adverse selection – MMing is not risk-free arbitrage.
  Rewards are funded by the platform's market-maker incentive program (from trading fees / grants), not taken from other traders.

$$\Pi_{MM} = \text{Spread} + \text{Rewards} - \text{Inventory Risk} - \text{Adverse Selection}$$

# Information reaches prices through order flow

**Private Info → Trade → Order Flow → Quote Rev. → Price Change**

- Informed traders hold a private signal and act on it.

- Their trading pressure moves the quote; the market maker re-prices.

**Illustrative example**

If an informed trader believes the true probability is 80% but the market is at 60%, they keep buying YES; order flow pushes the quote toward 80%. Information itself is invisible, but information-driven trading is observable – that is how prediction markets achieve price discovery.

# Price reflects more than average beliefs

- When YES = 0.70, the rigorous reading is: the market is willing to trade at ≈70% implied probability – not "everyone believes 70%."
- Beyond information, price is shaped by capital, risk appetite, positions, liquidity, fees, MM inventory, and subsidies.
- Empirically, treat the price as a "weight-adjusted willingness to trade", not an unbiased probability estimate.

1. Capital – large orders move price
2. Risk appetite – averse traders depress tails
3. Liquidity – thin books amplify noise
4. Inventory – MM hedging effects
5. Fees / Subsidies
6. Positions – e.g. a trader hedging an election book may sell YES on a correlated market

# Seven participants, seven sources of value

| Participant | Source of profit |
|---|---|
| Directional trader | more accurate forecast → buy low, sell high |
| Informed trader | information ahead of price, front-run repricing |
| Market maker | spread + rebates + liquidity incentives (net of risk) |
| Arbitrageur | YES+NO deviates from $1, or cross-market mispricing |
| Hedger | reduces risk in other positions, not necessarily profitable alone |
| Resolver / Oracle | rewards for correct or challenged resolution – proposers and disputers post a bond; correct calls win the bond, wrong ones are slashed |
| Platform | infrastructure, some fees, and data value – earns a share of trading fees and sells aggregated market data |

# Resolution determines which token pays $1

- After the outcome is clear, the market resolves: winning tokens redeem $1, losers go to 0.
- Uses UMA Optimistic Oracle: anyone can propose a result, and anyone can dispute an incorrect one.
  Proposers and disputers post a bond; if disputed, UMA token-holders vote. False proposals are slashed and honest disputers rewarded – the bond + vote blocks endless or abusive challenges.
- "Optimistic" means a result passes by default unless challenged within the dispute window – balancing efficiency and correction.
- On-chain settlement + disputability make payouts transparent and auditable, unlike a bookmaker.

**Resolution steps**

1. Outcome determined (real-world state settles) → 2. Submit result (anyone; bond posted, oracle accepts by default) → 3. Dispute window (can be challenged → token-holder vote) → 4. YES=1 / NO=0 (winners get $1, losers 0)

# From forecast performance to market mechanisms

Research increasingly asks which traders and institutions make prices informative.

- Early research establishes the foundations and conditions for information aggregation.
- Recent empirical work connects trader composition with profitability, liquidity provision, and forecast accuracy.
- A central unresolved question is whether changes in participation cause changes in market quality.

> This review identifies potential contribution areas without committing to a particular research direction.

# Information must be acquired, then aggregated

Information must first be acquired and then incorporated into prices.

- Grossman and Stiglitz (1980): Fully revealing prices would eliminate the incentive to acquire costly information. Equilibrium must preserve compensation for becoming informed.

- Ostrovsky (2012): Within a dynamic model of strategic trading, separable securities aggregate dispersed information under specified conditions.

- Together, these studies connect information incentives with the conditions governing price formation.

**Potential contribution**

Empirically connect changes in information costs with information acquisition, participation, and price informativeness.

# Trader composition may shape market accuracy

Aggregate market performance may depend heavily on a small group of participants.

- Heesen and Luong (2026) identify skilled traders using earlier markets and examine their participation in separate test markets.
- A greater trading share of skilled switchers is associated with lower pricing error, primarily through better calibration.
- Akey et al. (2026) document concentrated profits on Polymarket and find that successful users more often provide liquidity.

> **Interpretation**
>
> Who participates may matter alongside how much trading occurs.

# Activity, consensus, and accuracy are distinct

Each outcome captures a different aspect of market performance.

- Trading volume measures activity.
- Liquidity measures describe trading conditions.
- Transaction-price dispersion captures differences in execution prices within a measurement window.
- Calibration evaluates whether realized event frequencies correspond to quoted probabilities.

**Page and Clemen (2013)**

document horizon-dependent calibration and systematic favorite–longshot bias.

**Potential contribution**

Explain when improvements in trading conditions or price consensus translate into better-calibrated forecasts.

# The central empirical gap is causal identification

Existing studies describe trader behavior more clearly than they identify its effects.

- Skilled traders may choose markets that are already easier to price accurately.
- Identifying skill in earlier markets reduces circularity but does not eliminate market selection.
- Liquidity providers' profits may reflect execution advantages or compensation for risk, in addition to information advantages.
- These findings therefore do not establish that increasing skilled participation or maker activity causes better forecasts.

**Potential contribution**

Identify a credible source of variation in participation and estimate its effect on market quality.

References: Akey et al. (2026); Heesen and Luong (2026).

# How do information costs shape participation?

How do changes in information costs affect who acquires information and who trades?

- Lower costs may allow additional participants to become informed.
- Changes in relative information advantages may alter entry, exit, and trading intensity.
- The resulting participation changes may affect both information production and liquidity supply.
- AI is one possible setting. Other candidates include changes in data access or information disclosure, provided they offer credible identification.

> **Evidence needed**
>
> A measurable change in information costs and a design that separates its effects from concurrent changes.

References: Theoretical motivation: Grossman and Stiglitz (1980).

# Which trading functions improve accuracy?

Which trading functions account for the relationship between skilled participation and accurate prices?

- Skilled traders may contribute through aggressive trades that incorporate information into prices.
- They may also supply liquidity that allows other participants to trade.
- Execution advantages, inventory risk, and information advantages may explain different components of profitability.
- Persistent liquidity provision may matter differently from occasionally submitting a limit order.

> **Evidence needed**
>
> Measures that distinguish forecasting skill, execution role, and sustained liquidity provision, ideally combined with independent variation in participation.

References: Akey et al. (2026); Heesen and Luong (2026).

# When does price consensus signal better information?

More similar prices can arise from different underlying mechanisms.

- Trading may successfully incorporate dispersed information.
- Participants may instead rely on common signals and share similar errors.
- Changes in the composition or timing of trades may also reduce measured price dispersion.

**Potential contribution**

Distinguish these mechanisms and establish when stronger price consensus predicts better calibration.

**Evidence needed**

Transaction data linked to realized outcomes, with comparable prediction horizons and attention to changes in trading composition.

References: Ostrovsky (2012); Page and Clemen (2013).

# Four directions, distinct identification needs

The reviewed literature suggests several promising questions, but does not yet determine a project.

| Potential direction | Central question | Main empirical requirement |
|---|---|---|
| Information costs | How do cost changes affect information acquisition and participation? | Credible variation in information costs |
| Trader composition | Does skilled participation improve market quality? | Variation that addresses market selection |
| Liquidity provision | Which trading functions explain profitability and accuracy? | Distinct measures of skill, roles, and risks |
| Price consensus | When does convergence in prices indicate better forecasts? | Outcome-based calibration tests |

- The next step is to assess novelty, data feasibility, and identification for each direction.

- These are potential contribution areas suggested by the reviewed papers. Establishing an unfilled gap requires checking the broader literature.

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